



Basic Principles of Medicine 1
Module: Musculoskeletal System
DLA

Somatic Nerves and Plexuses

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Recommended Reading

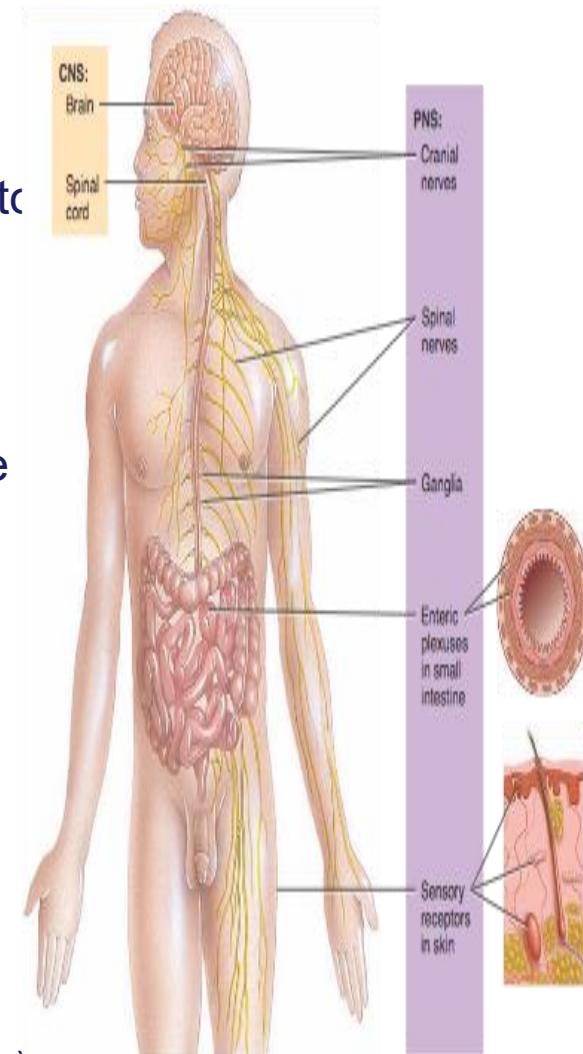
- Drake, Vogl, Mitchell. Gray's Anatomy for students. 4th ed. Elsevier.
- Printed Textbook Chapter 2 pg 109-113
- Online Textbook Chapter 2:
 - Spinal cord
- Pay special attention to the green “In the Clinic” boxes for the clinical correlations for this topic
- Image references
- Torso. Waschke, Jens, Sobotta Anatomy Textbook, 3, 75-137.
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Objectives

SOM.MKII.BPM1.1.MSK.1.ANAT.1064	Define the terms somatic afferent, visceral afferent, somatic efferent, and visceral efferent as they apply to nerve fibers found in spinal nerves.
SOM.MKII.BPM1.1.MSK.1.ANAT.1065	List the different modalities and targets of the fibers found in spinal nerve
SOM.MKII.BPM1.1.MSK.1.ANAT.1066	Compare and contrast differences between dorsal and ventral roots, compare the contrast differences between dorsal and ventral rami.
SOM.MKII.BPM1.1.MSK.1.ANAT.1067	Describe a dorsal root ganglion and the pathway of a typical afferent neuron.
SOM.MKII.BPM1.1.MSK.1.ANAT.1068	Define a dermatome, myotome and discuss the clinical importance of understanding these concepts.
SOM.MKII.BPM1.1.MSK.1.ANAT.1069	Describe which dermatomes are associated with the following anatomical landmarks/structures: lateral border of upper limb, thumb, medial border of upper limb, little finger, nipple, xiphoid, umbilicus, pubic region, big toe, little toe.
SOM.MKII.BPM1.1.MSK.1.ANAT.1070	Explain how the spinal nerves exit the five regions of the vertebral column and use this information to interpret clinical signs and symptoms
SOM.MKII.BPM1.1.MSK.1.ANAT.1071	Describe the organization and composition of a somatic nerve plexus and list the nerve fiber types found in each component.
SOM.MKII.BPM1.1.MSK.1.ANAT.1072	Outline the dermatome map of the body and use it to explain the reappearance of herpes zoster (shingles) in a particular dermatome(s).
SOM.MKII.BPM1.1.MSK.1.ANAT.1073	Describe the pathways of sympathetic fibers innervating the body wall
SOM.MKII.BPM1.1.MSK.1.ANAT.1074	Describe the course of visceral sensory fibers traveling with sympathetic nerves, and the location of their cell bodies.
SOM.MKII.BPM1.1.MSK.1.ANAT.1075	Describe the mechanism of referred pain.

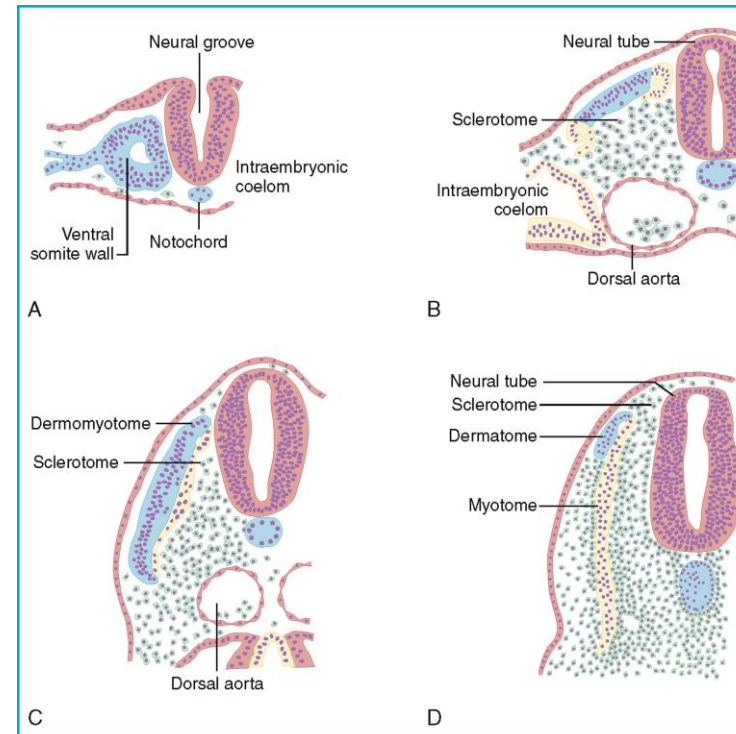
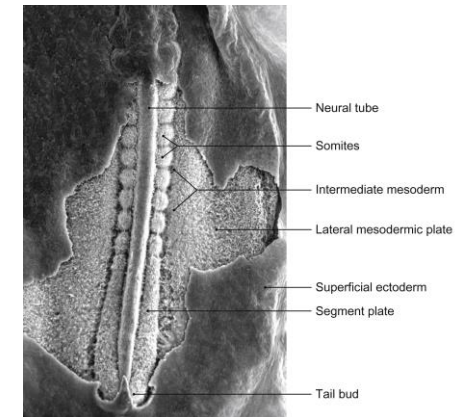
Overview – summary and recap from FTM

- Central nervous system (CNS)
 - Brain (cerebrum, cerebellum, brainstem and diencephalon) and spinal cord
 - Receives information from and sends information to the body, serves as the central control unit
- Peripheral nervous system (PNS)
 - Everything else
 - The pathways (nerves) that communicate from the body structures to the CNS and vice versa
- Autonomic nervous system (CNS & PNS)
 - The 'automatic' control of the body
 - Contains both central and peripheral components
- Other divisions
 - Somatic
 - Relating to the body (muscles, skin, bones)
 - Visceral
 - Relating to the organs (heart, vessels, digestive etc.)



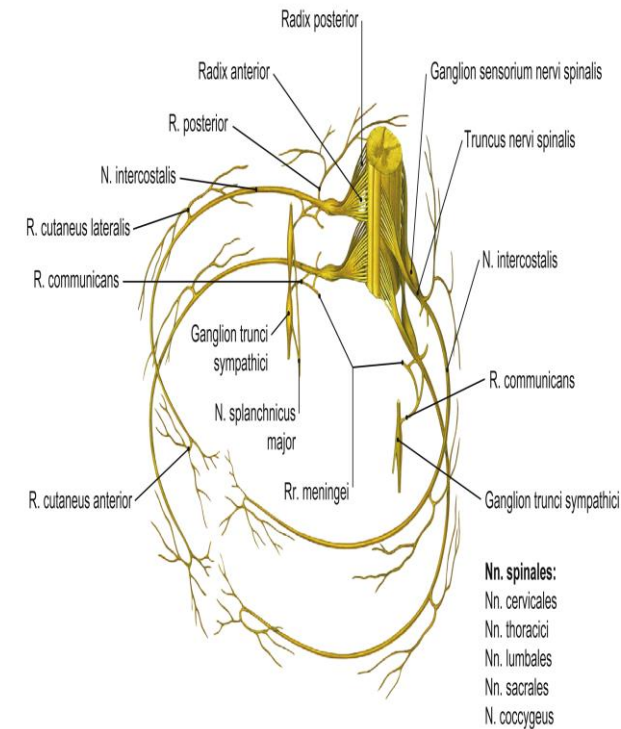
Somites

- The vertebral column, skin and skeletal muscles of the body and limbs develop embryologically from somites
- Somites form from the paraxial mesoderm
- They develop alongside the developing neural tube as paired segments
- Two major parts
 - Sclerotome – vertebral column
 - Dermomyotome
 - Dermatome - skin
 - Myotome – muscles of trunk and limbs
- General concept is that the dermatome and myotome develop from a single segment and is innervated by a single spinal nerve

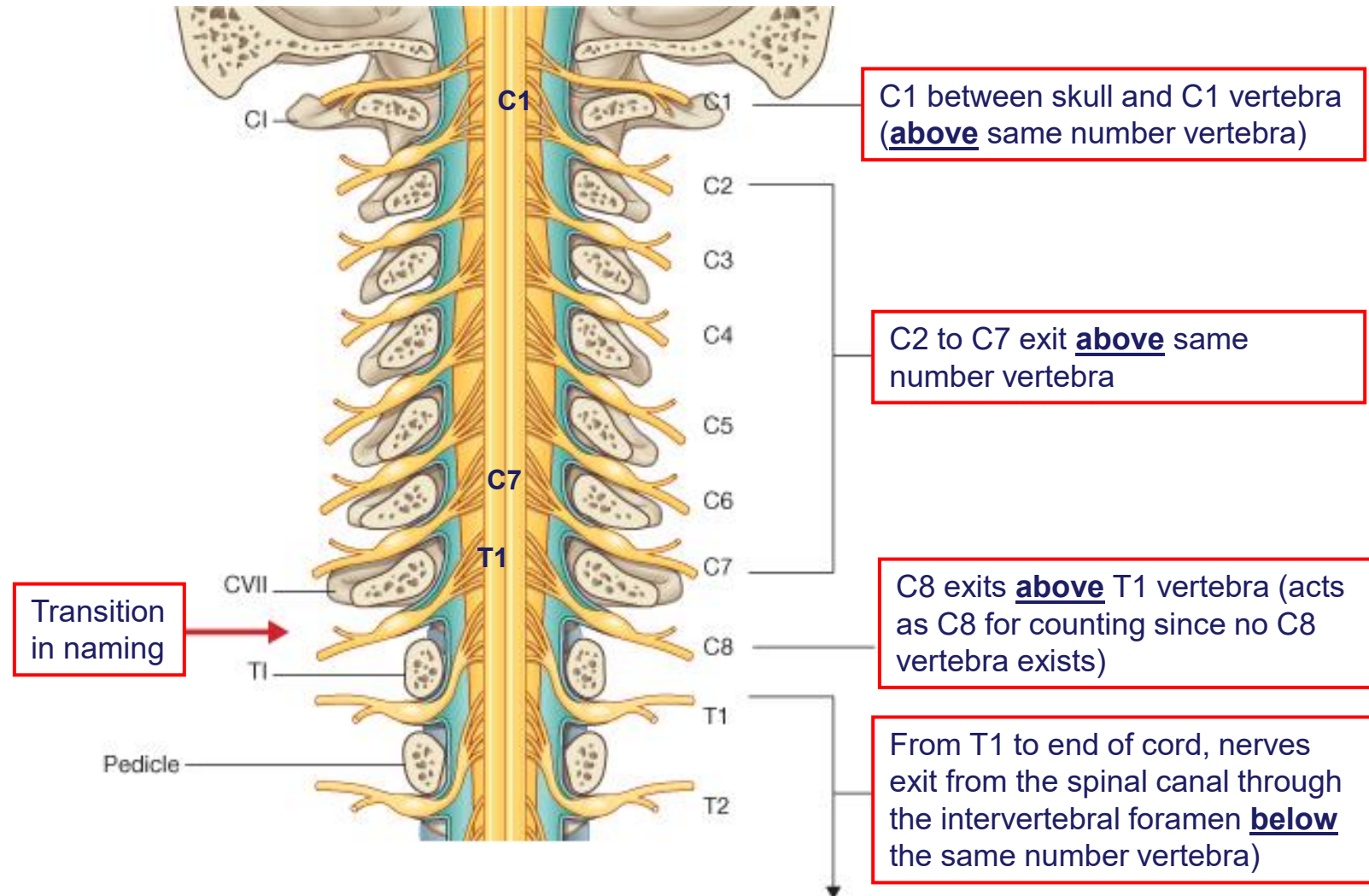


Typical spinal nerve course

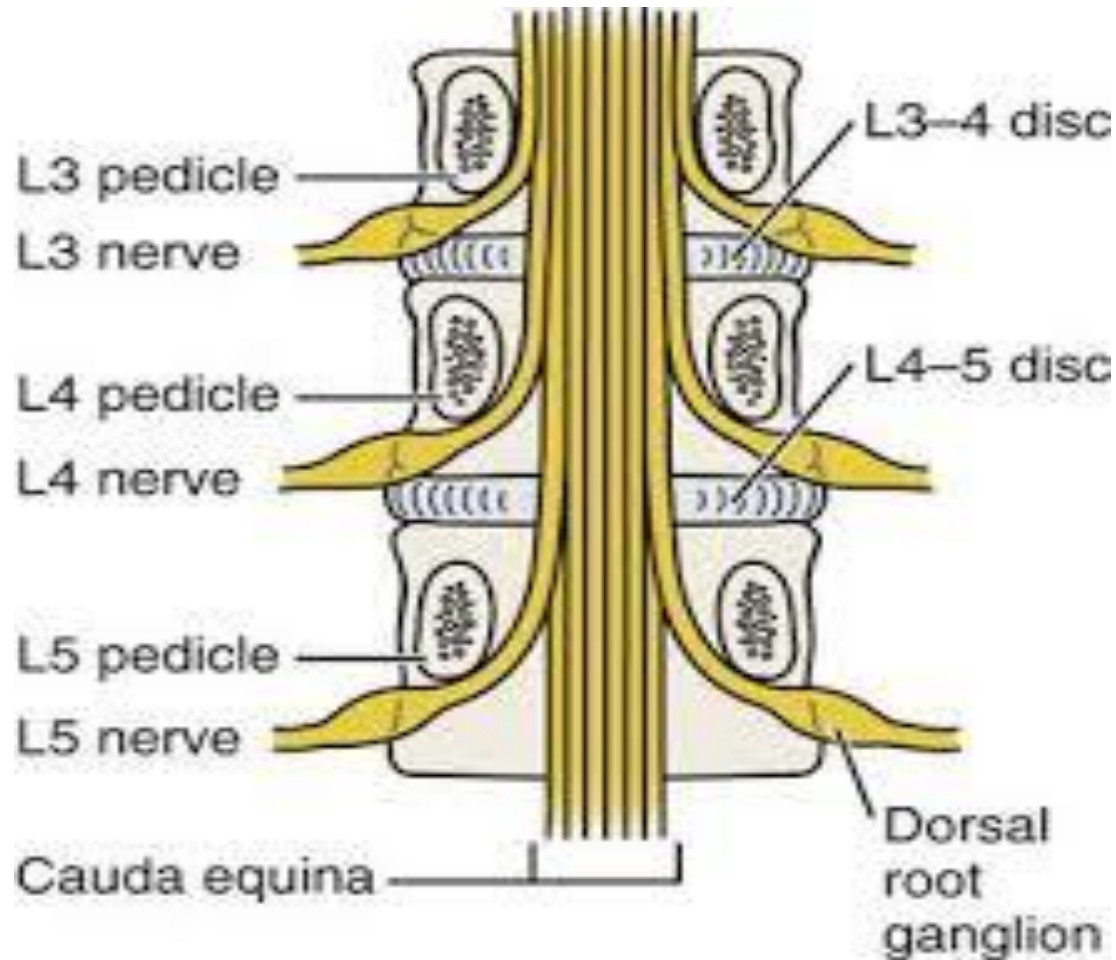
- Exits the intervertebral foramen as a mixed, motor and sensory nerve
- Dorsal (Posterior) ramus – supplies intrinsic back muscles (motor) and skin over that area (sensory)
 - Also contains the postganglionic sympathetic fibers that innervate the skin glands and vessels of the back
- Ventral (Anterior) ramus - supplies all other muscles (motor) and skin over that area (sensory)
 - At levels T1 – L2 it contains the preganglionic sympathetic fibers
 - At all levels it contains the postganglionic sympathetic fibers that innervate the skin glands and vessels of the area supplied by the nerve
 - Also receives visceral afferents from organs at levels T1-L2 via the white rami communicans



Spinal Nerves and Their Path

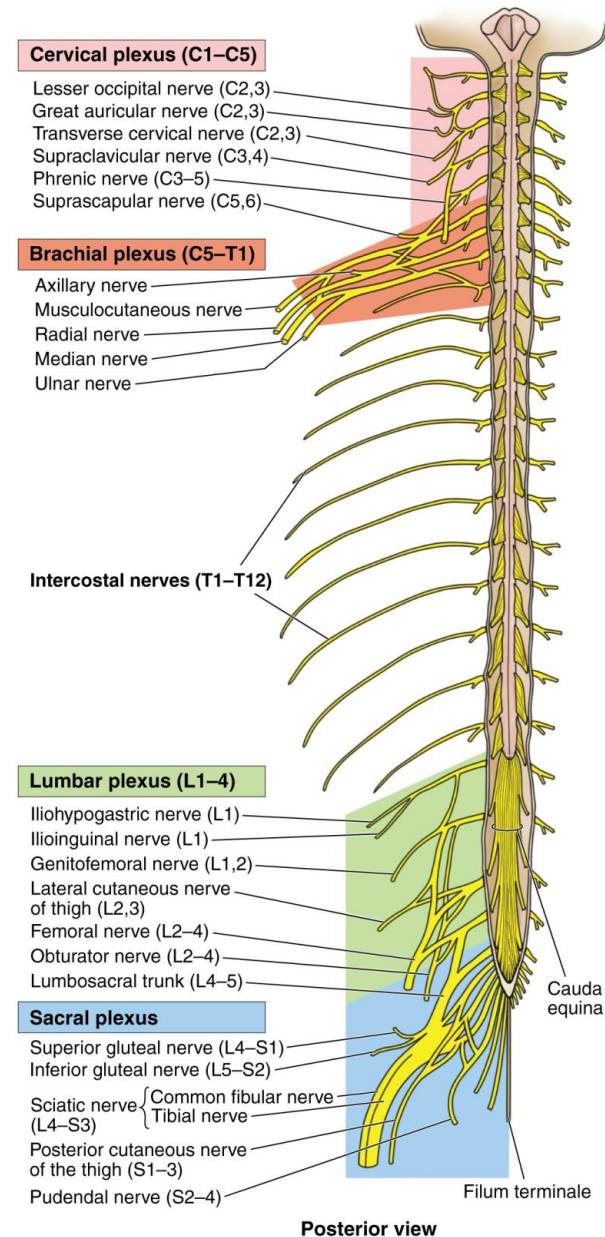


Spinal nerve exit in lumbar region



Organization of peripheral nerves

- Muscles form from multiple myotomes, because each myotome has its own spinal nerve.
- Plexus - A network of intersecting nerves from different levels or sources that combine to form new nerves with specific targets
- Somatic plexuses: Form from the ventral rami of spinal nerves
 - C1 – 5 – Cervical plexus
 - C5 – T1 – Brachial plexus
 - L1 – 4 – Lumbar plexus
 - L5 – S4 – Sacral plexus
- Combination of somatic sensory, somatic motor and sympathetic nerve fibers
- Visceral plexuses
 - Combination of sympathetic, parasympathetic and visceral afferent fibers
 - e.g. hypogastric, celiac



General concept that applies to afferent (sensory) fibers

The general arrangement is a three neuron chain.

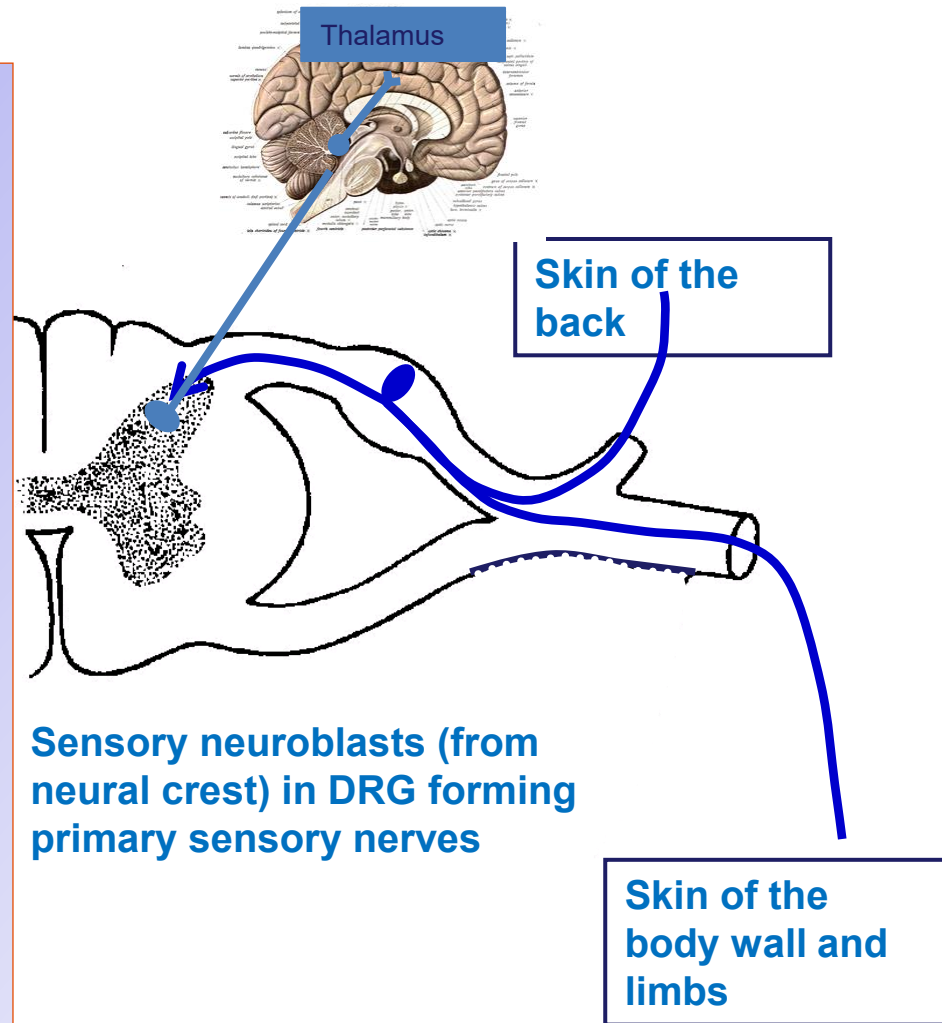
The first (1st order – 1°) arises at the organ (or muscle) and has two processes, one on either side of the cell body.

The cell body is in a ganglion located outside of the CNS (**no synapse occurs here**).

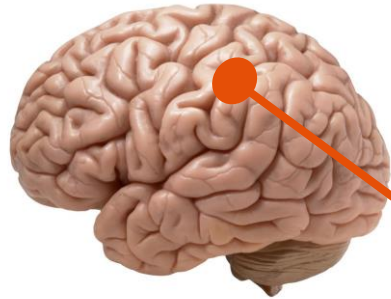
The central process enters the CNS (spinal cord/brainstem) synapses on the next neuron in the chain.

The second (2nd order – 2°) arises from the grey horn of the spinal cord or brainstem nucleus and synapses in the thalamus.

The third (3rd order) arises from the thalamus and ends in the cortex.



General concept that applies to efferent (motor) fibers



The general arrangement is a two-neuron chain

The first (upper motor neuron - UMN) arises in the cortex and synapses in the ventral horn of the spinal cord or in the brainstem on the nucleus of the corresponding cranial nerve.

The second (lower motor neuron – LMN) arises from the brainstem nucleus and ends in the muscle it supplies

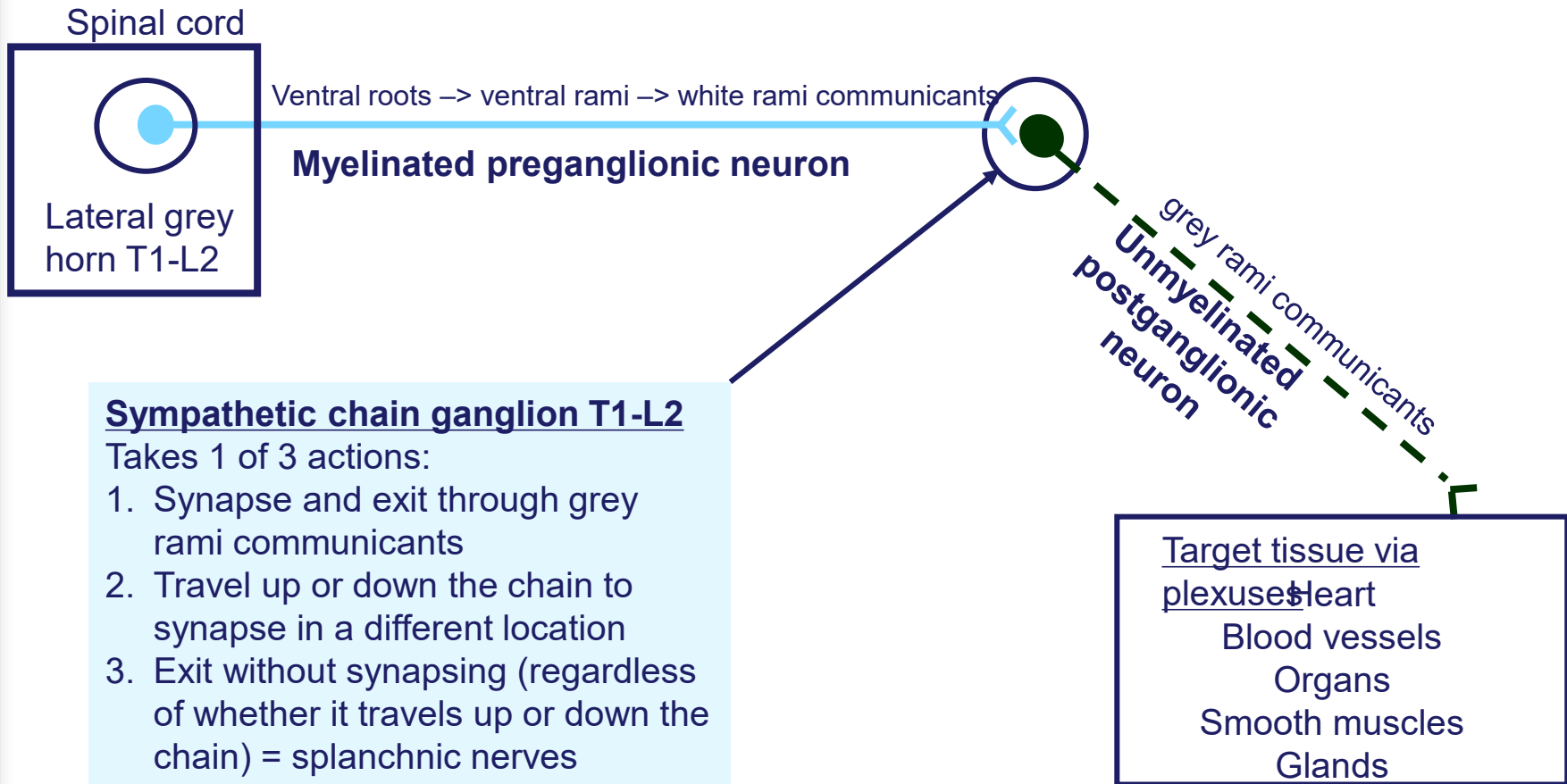


Motor neuroblast (from basal plate of intermediate layer) forming somatic motor neuron

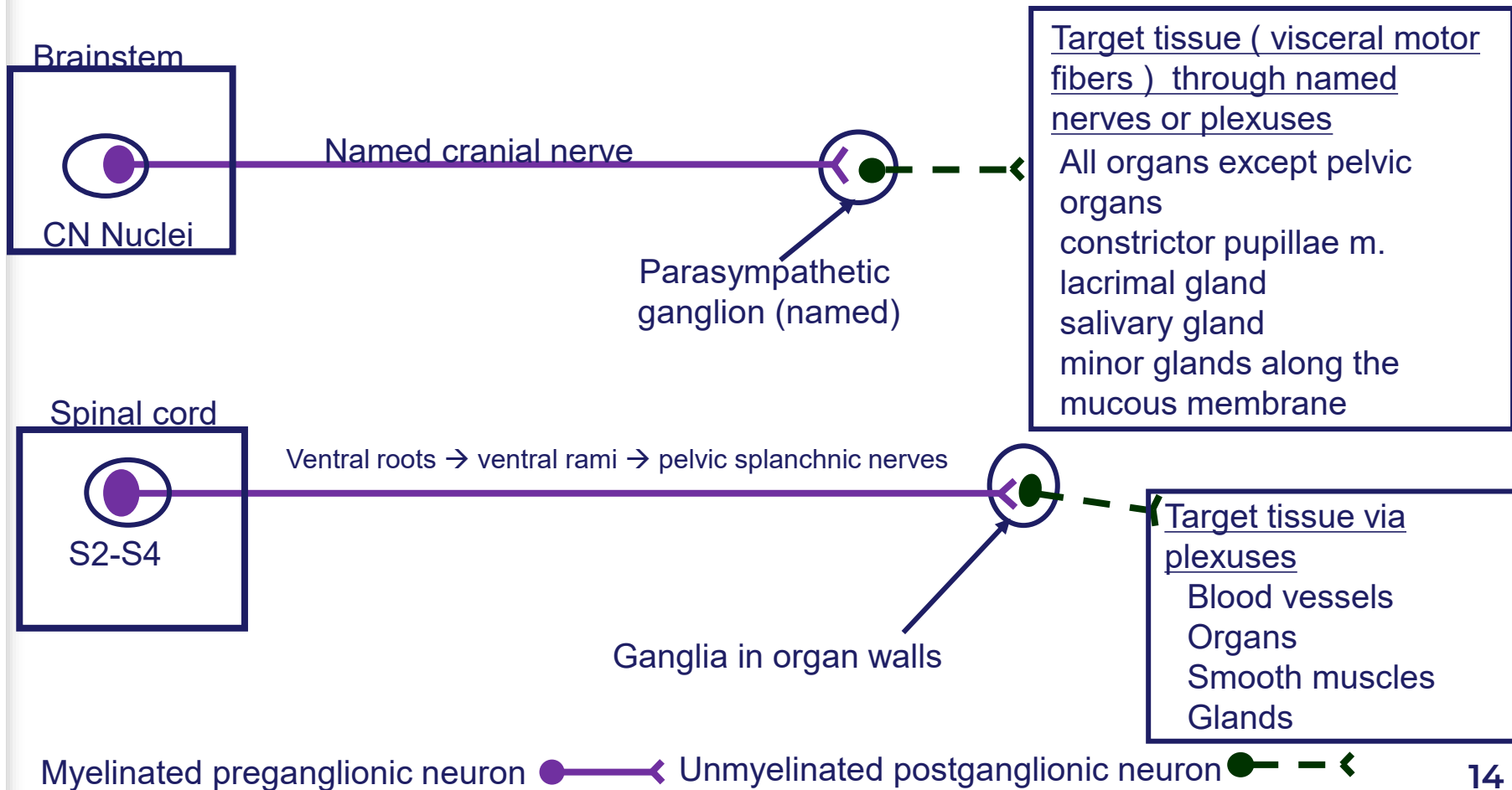
Muscles of the back

Muscles of the body wall and limbs

General concept that applies to sympathetic efferent fibers

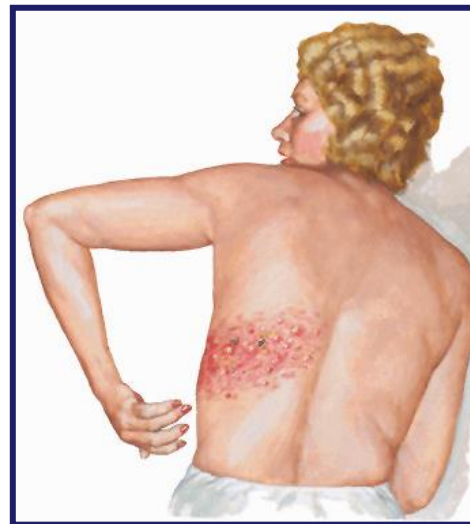


General concept that applies to parasympathetic efferent fibers

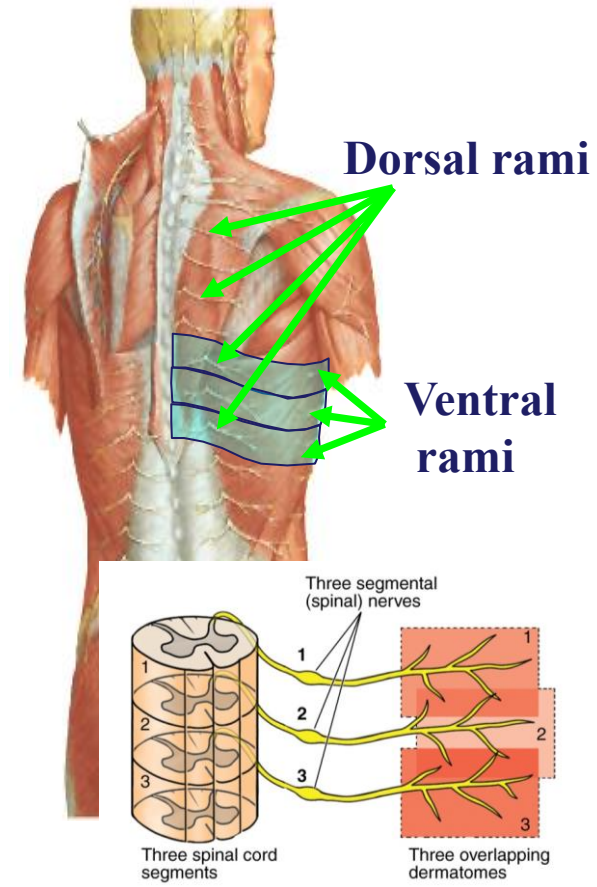


Dermatomes

- An area of skin supplied by a single spinal nerve
- There is a degree of overlap between the dermatomes
- Historically there are 3 maps – we use a combination
- Understanding the dermatomes not only allows us to pinpoint a lesion but also links to the concept of referred pain
- Some clinically important dermatomes
 - C5 – Lateral upper arm
 - C6 – thumb
 - C7 – middle finger
 - C8 – little finger
 - T2 – medial arm
 - T4 – nipple
 - T7 – xiphoid
 - T10 – umbilicus
 - L4 – big toe
 - S1 – heel
 - S2 – back of thigh

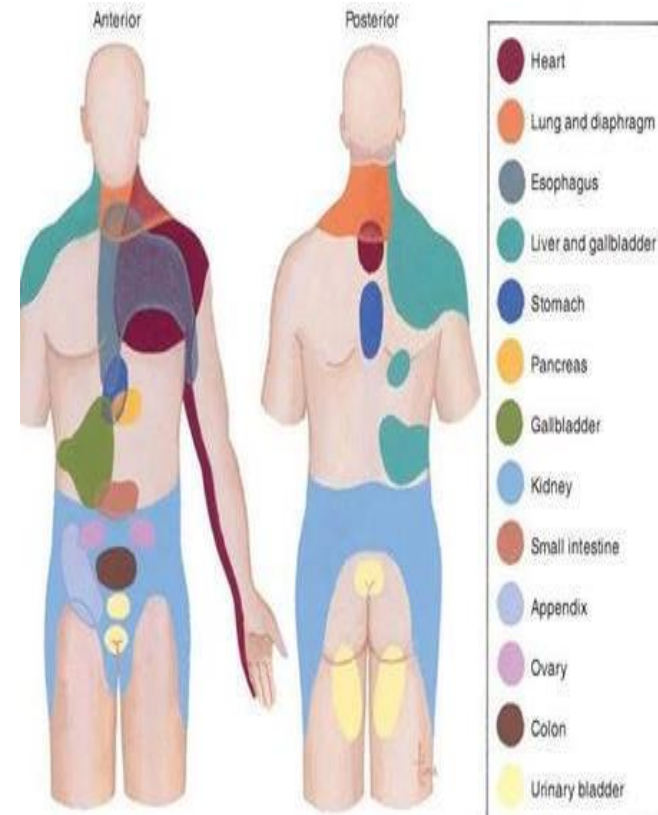


Shingles – herpes zoster



Referred pain

- When pain is felt in an area that is different to where the injury occurred
 - This often happens with organ injury which may be felt on the skin of a limb or the wall of the abdomen
- There are different types of referred pain and knowledge of where they refer to is important for clinical diagnosis
- It is not a well understood phenomenon, but it has been linked to many possible reasons
 - The visceral pain fibers synapse on the same 2nd order neurons as the somatic pain fibers
 - Visceral pain fibers are lower threshold and exceed the somatic fibers in number by a factor of 10
- This leads to the pain being felt on the skin
- The referred pain patterns of each organ is predictable and well known



Nervous System summary

